

FORT COLLINS, CO, USA

WMO: 724697

Lat: 40.588N

Lon: 105.042W

Elev: 4939

StdP: 12.26

Time Zone: -7.00 (NAM)

Period: 86-94

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-2.6	4.8	-12.5	3.4	3.8	-5.2	5.0	9.9	24.7	45.8	19.7	43.4	4.7	20	0.515

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	24.5	90.1	60.9	87.2	60.4	84.4	60.1	64.6	80.7	63.6	80.0	62.5	78.6	8.3	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
59.7	92.1	69.7	58.4	87.9	69.2	57.0	83.4	68.7	32.5	81.3	31.6	79.2	30.7	78.9	74.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.0	16.8	13.6	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
			-8.8	68.7	8.5	2.7	-15.0	70.6	-20.0	72.2	-24.7	73.7	-30.9	75.7

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	49.6	30.1	32.9	40.9	49.2	58.0	67.0	70.6	68.7	61.3	49.8	37.2	28.1
	DBStd	17.02	9.59	11.23	9.08	8.80	6.28	5.77	4.48	4.58	6.09	8.67	9.48	10.76
	HDD50	2707	619	482	296	118	12	0	0	0	5	105	390	680
	HDD65	6096	1083	900	747	476	226	45	9	22	139	470	834	1145
	CDD50	2550	1	2	15	94	260	511	639	578	343	101	6	0
	CDD65	462	0	0	0	2	8	106	183	135	27	1	0	0
	CDH74	5659	0	0	2	88	272	1315	1979	1416	498	89	0	0
	CDH80	1787	0	0	0	12	49	471	708	453	85	9	0	0
(14) Wind	WSAvg	6.3	6.1	6.4	7.2	7.4	7.0	6.5	6.1	5.8	5.9	5.7	6.2	5.6
(15) Precipitation	PrecAvg	15.5	0.4	0.4	1.4	1.9	2.6	2.0	1.7	1.4	1.3	1.1	0.7	0.4
	PrecMax	28.4	1.2	1.3	5.7	8.3	7.5	5.8	6.7	5.1	4.0	4.8	2.3	1.5
	PrecMin	7.3	0.0	0.0	0.0	0.3	0.0	0.1	0.1	0.0	0.0	0.1	0.0	0.0
	PrecStd	4.3	0.3	0.3	1.3	1.3	1.7	1.3	1.4	1.1	1.0	0.9	0.6	0.4
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	60.3	66.2	72.0	81.3	84.7	93.5	93.6	91.2	86.4	81.2	68.0	61.2
		MCWB	41.5	45.7	47.7	53.9	55.6	61.3	60.7	60.0	58.4	52.9	46.7	42.8
	2%	DB	54.5	58.8	67.3	76.1	80.7	88.7	90.5	88.1	82.6	76.8	62.5	54.8
		MCWB	38.9	42.1	45.9	52.1	55.7	60.3	61.8	60.0	57.9	52.6	45.5	39.9
	5%	DB	50.1	54.5	63.1	72.2	77.0	85.7	87.6	85.1	79.7	72.5	57.3	49.0
		MCWB	37.1	39.9	44.1	51.0	54.8	59.4	61.5	60.6	57.1	50.2	43.0	36.1
	10%	DB	45.7	49.9	58.2	66.9	73.4	82.2	84.4	81.6	76.6	67.4	52.8	43.7
		MCWB	34.4	37.3	42.5	49.1	54.4	59.2	61.3	60.4	56.7	48.8	40.3	33.1
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	42.8	46.6	49.4	56.8	59.5	65.7	66.9	65.4	63.8	56.0	49.4	44.4
		MCDB	59.5	64.3	68.0	77.8	76.7	83.6	81.9	78.6	79.0	75.9	61.7	58.1
	2%	WB	39.7	42.8	46.9	53.7	58.0	63.8	65.0	64.2	61.2	53.0	46.4	40.5
		MCDB	53.8	57.1	65.4	71.6	73.8	81.1	81.8	78.2	75.1	74.1	60.5	54.0
	5%	WB	37.1	40.4	44.9	51.9	56.6	62.2	63.8	63.2	59.4	51.1	43.6	36.6
		MCDB	49.5	53.6	61.0	69.3	72.3	78.8	81.5	77.6	74.0	69.1	56.7	48.0
	10%	WB	34.8	37.7	43.0	49.8	55.1	60.7	62.6	62.2	57.8	49.6	41.1	33.3
		MCDB	45.6	48.9	57.6	66.3	70.8	76.9	80.3	76.6	72.7	66.1	51.7	43.2
(35) Mean Daily Temperature Range	5% DB	MDBR	22.6	22.1	22.8	23.9	23.4	24.3	24.5	23.5	25.5	25.4	21.8	22.2
		MCDBR	29.0	30.3	33.6	32.9	31.6	31.0	30.4	30.6	31.9	35.4	29.6	30.7
		MCWBR	17.0	17.1	17.6	15.2	13.5	11.1	11.2	10.6	13.0	16.1	16.3	18.4
	5% WB	MCDBR	27.8	29.4	31.0	29.3	26.6	26.2	25.3	23.0	26.7	31.3	29.6	29.5
		MCWBR	16.6	16.9	17.4	15.0	12.3	11.0	10.8	9.8	12.2	15.4	16.8	18.3
(40) Clear-Sky Solar Irradiance	taub		0.245	0.254	0.279	0.305	0.329	0.336	0.366	0.360	0.316	0.278	0.259	0.249
	taud		2.583	2.566	2.507	2.468	2.459	2.456	2.416	2.422	2.528	2.610	2.574	2.566
	Ebn at Noon		301	312	312	308	300	297	287	286	294	298	292	290
	Edh at Noon		21	25	30	34	35	35	36	35	30	24	21	20
(44) All-Sky Solar Radiation	RadAvg		779	1087	1479	1728	1916	2256	2133	1874	1613	1157	822	660
	RadStd		37	67	115	118	161	122	88	69	92	80	51	45

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (27 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page